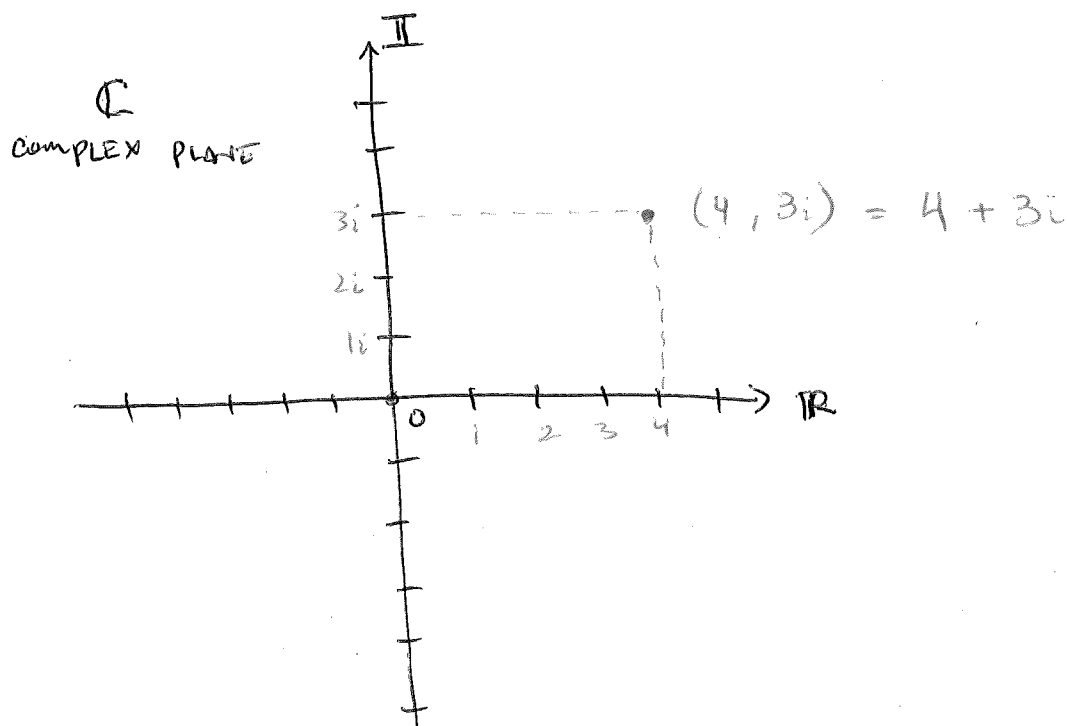




ANY COMPLEX $z = x + iy$

COMPLEX CONJUGATE (*)

$$z^* = x - iy$$

↑
CHANGES SIGN

e.g. $z = 4 + 3i$, $z^* = 4 - 3i$

$z = -4 - 3i$, $z^* = -4 + 3i$

$\text{Re}(z)$, $z = x + iy \Rightarrow \text{Re}(z) = x$

$$\text{Re}(z) = \frac{z + z^*}{2}$$

e.g. $z = 4 + 3i$

$$\begin{aligned} \text{Re}(z) &= \frac{(4 + 3i) + (4 + 3i)^*}{2} \\ &= \frac{4 + 3i + 4 - 3i}{2} \\ &= \frac{8}{2} = 4 \end{aligned}$$

$$\operatorname{Im}(z), \quad z = x + yi \quad \operatorname{Re}(z) = x$$

③

$$\operatorname{Im}(z) = \frac{z - z^*}{2i}$$

eg. $z = 4 + 3i$

$$\operatorname{Im}(z) = \frac{(4 + 3i) - (4 + 3i)^*}{2i}$$

$$= \frac{4 + 3i - (4 - 3i)}{2i}$$

$$= \frac{6i}{2i}$$

$$= 3$$

ADD

$$z_1 + z_2 =$$

$$z_1 = (a + bi) \quad z_2 = (c + di)$$

$$= (a + bi) + (c + di)$$

$$= (a + c) + (b + d)i$$

$$z_1 + z_2 = z_2 + z_1$$

MULTIPLICATION

$$z_1 z_2 = (a + bi)(c + di)$$

$$= ac + adi + bci + bdi^2$$

$$= ac + adi + bci - bd$$

$$= (ac - bd) + (ad + bc)i$$

$$z_1 z_2 = z_2 z_1$$

Division

$$\frac{z_1}{z_2} = \frac{(a+bi)}{(c+di)}$$

$$z = x + yi$$

$$\frac{z_1 z_2^*}{z_2 z_2^*} = \frac{a+bi}{c+di} \frac{(c-di)}{(c-di)}$$

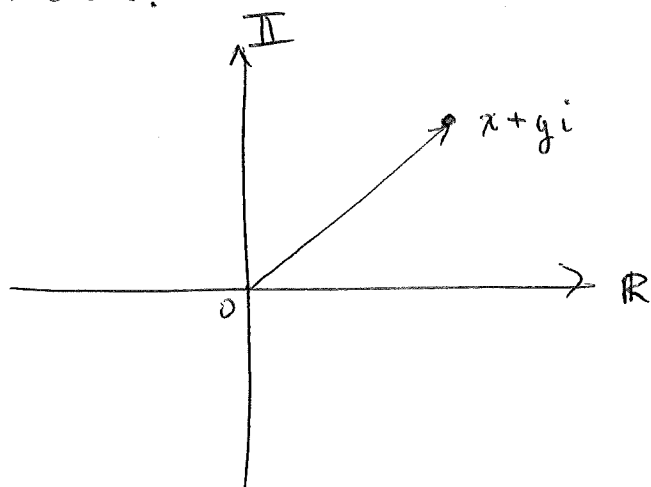
$$= \frac{(ac+bd) + (bc-ad)i}{c^2 + d^2}$$

$$= \left(\frac{ac+bd}{c^2+d^2} \right) + \frac{(bc-ad)i}{c^2+d^2}$$

real

imaginary

Norm.



vector

$$\# |v| = (v \cdot v)^{1/2}$$

correct

$$|z| = (z \cdot z)^{1/2}$$

wrong

$$((x+yi)(x+yi))^{1/2}$$

$$(x^2 - y^2 + 2xyi)^{1/2}$$

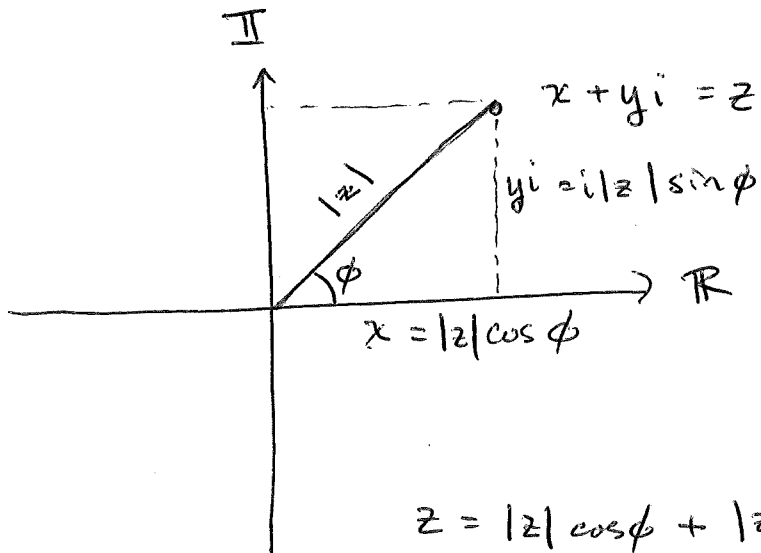
$$|z| = (z \cdot z)^{1/2} = (zz^*)^{1/2}$$

$$= ((x+yi)(x+yi)^*)^{1/2}$$

$$= ((x+yi)(x-yi))^{1/2}$$

$$\text{MAGNITUDE } |z| = (x^2 + y^2)^{1/2}$$

(4)



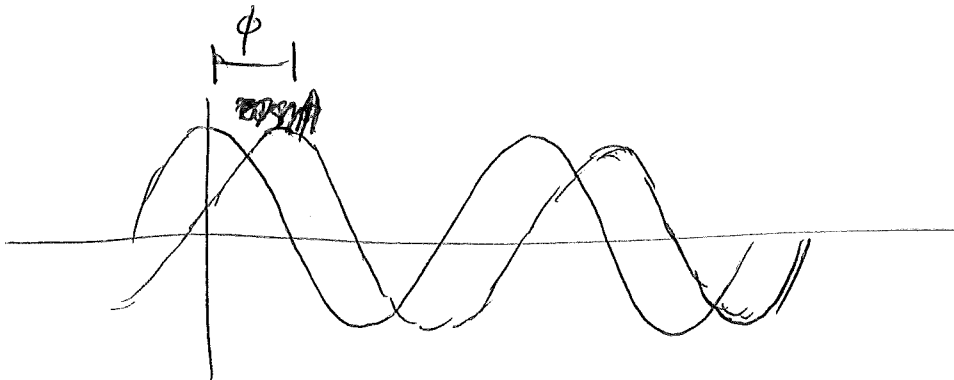
$$z = |z| \cos \phi + |z| i \sin \phi$$

$$= |z| (\cos \phi + i \sin \phi)$$

$$|z| = (x^2 + y^2)^{1/2} \quad \text{--- AMPLITUDE}$$

$$\phi = \tan^{-1} \frac{\text{Im}(z)}{\text{Re}(z)}$$

$$\phi = \tan^{-1} \frac{y}{x} \quad \text{--- PHASE}$$



e.g. $z^* = (|z|(\cos \phi + i \sin \phi))^*$

$$= |z|(\cos \phi - i \sin \phi)$$

5

$$\cos \phi = 1 - \frac{\phi^2}{2!} + \frac{\phi^4}{4!} - \dots$$

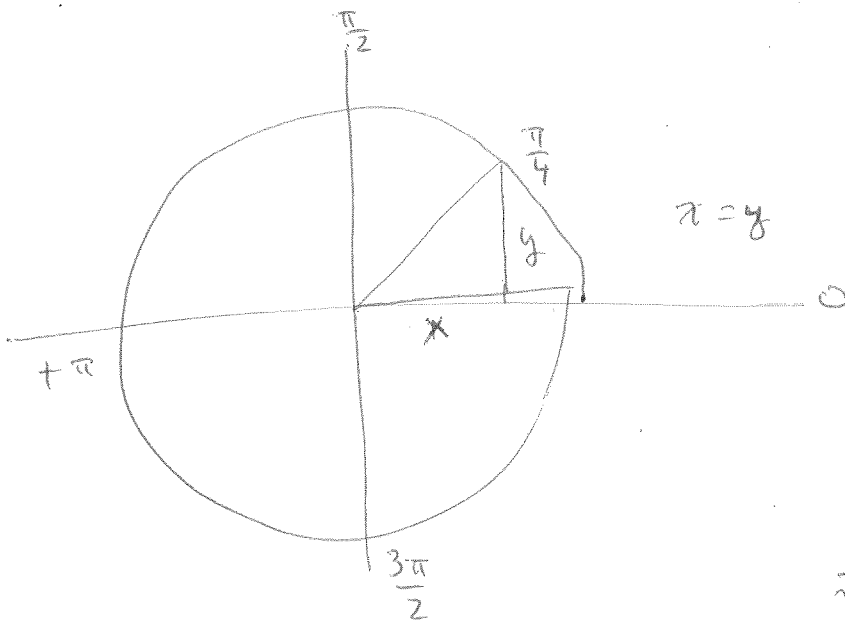
$$i \sin \phi = i\phi - \frac{i\phi^3}{3!} + \frac{i\phi^5}{5!} - \dots$$

$$\cos \phi + i \sin \phi = 1 + i\phi - \frac{\phi^2}{2!} - \frac{i\phi^3}{3!} + \frac{\phi^4}{4!} + \frac{i\phi^5}{5!} - \dots$$

$$e^{i\phi} = \cos \phi + i \sin \phi$$

eg. $e^{i0} = e^0 = 1$

$$e^{i\frac{\pi}{4}} = \cos \frac{\pi}{4} + i \sin \frac{\pi}{4}$$



$$1^2 = x^2 + y^2$$

$$1^2 = x^2 + x^2$$

$$1^2 = 2x^2$$

$$\frac{1}{2} = x^2$$

$$\sqrt{\frac{1}{2}} = x$$

$$\frac{1}{\sqrt{2}} \cdot \frac{\sqrt{2}}{\sqrt{2}} =$$

$$\frac{\sqrt{2}}{2} =$$

$$e^{i\frac{\pi}{4}} = \frac{\sqrt{2}}{2} + i \frac{\sqrt{2}}{2}$$

$$e^{i\frac{\pi}{2}} = 0 + i1 = i$$

$$e^{i\pi} = -1 + 0i$$

$$e^{i\pi} = -1$$

Euler's formula

(6)

$$z = |z| (\cos \phi + i \sin \phi)$$

$$z = |z| e^{i\phi}$$

TAYLOR SERIES

$$f(z) = f(a) + \sum_{n=1}^{\infty} f^{(n)}(a) \frac{(z-a)^n}{n!}$$

FACTORIAL $n! = 1 \cdot 2 \cdot 3 \cdot \dots \cdot n$

$$a=0, \quad f(z) = \exp(z) = e^z \quad z \in \mathbb{R}$$

$$e^z = 1 + z + \frac{z^2}{2} + \frac{z^3}{6} + \frac{z^4}{4!} + \frac{z^5}{5!} + \dots$$

$$\cos(z) = 1 - \frac{z^2}{2!} + \frac{z^4}{4!} - \dots$$

$$\sin(z) = z - \frac{z^3}{3!} + \frac{z^5}{5!} - \dots$$

$$e^{i\phi} = 1 + i\phi + \frac{(i\phi)^2}{2!} + \frac{(i\phi)^3}{3!} + \frac{(i\phi)^4}{4!} + \frac{(i\phi)^5}{5!} + \dots$$

$$i^0 = 1$$

$$i^1 = i$$

$$i^2 = -1$$

$$i^3 = -i$$

$$i^4 = 1$$

$$i^5 = i$$

$$e^{i\phi} = 1 + i\phi - \frac{\phi^2}{2!} - \frac{i\phi^3}{3!} + \frac{\phi^4}{4!} + \frac{i\phi^5}{5!} + \dots$$

(7)

$$e^{in\pi} \quad n; \text{ even} \quad 1$$

$$n \text{ odd} \quad -1$$

$$E_x(z) = E_0 \exp\left(-\frac{(1+i)}{\sqrt{2}} z\right) \quad z, \text{ depth}$$

$$= E_0 \exp\left(-\frac{1}{\sqrt{2}} z - \frac{i}{\sqrt{2}} z\right)$$

$$= E_0 \exp\left(-\frac{1}{\sqrt{2}} z\right) \exp\left(-\frac{i}{\sqrt{2}} z\right)$$

MAGNITUDE

PHASE

$$= E_0 \exp\left(-\frac{1}{\sqrt{2}} z\right) \left[\cos\left(-\frac{1}{\sqrt{2}} z\right) + i \sin\left(-\frac{1}{\sqrt{2}} z\right) \right]$$

$$E_x(z) = E_0 \exp\left(-\frac{1}{\sqrt{2}} z\right) \left[\cos\left(\frac{1}{\sqrt{2}} z\right) - i \sin\left(\frac{1}{\sqrt{2}} z\right) \right]$$

EXPONENTIAL
DECAY

CIRCLE